

Baud divider

Your system clock runs much faster than the UART bit rate

Starter DIV equals four

- Starter preset: DIV equals four in baud_tick pulse mode, sixteen sysclk cycles shown
- Count zero, one, two, three, then baud_tick equals one on the wrap at cnt equals three
- In sixteen cycles you see four ticks
- Step once to watch cnt advance, or play to end for the full pattern
- Compare with toggle mode

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Baud / clock divider

From **sysclk** to a **baud_tick** pulse (UART) or a toggled **clk_div** (SPI-style). Count 0 ...
DIV-1, fire on wrap. Starter: DIV=4 pulse — tick every 4 clocks.

Starter example: DIV=4, **baud_tick** pulse mode, 16 sysclk cycles. Tick when cnt==3, then wrap to 0. Compare with toggle mode (SPI-style). Use the calculator for real f_{sys} / baud values.

Load starter example

Challenges 0 / 22 cleared

Quiz: DIV: A common integer baud divider is...

☐ DIV $\approx$ round(f_{sys} / baud)

☐ DIV = baud / f_{sys} always

☐ DIV must be a power of two only

☐ DIV is the UART stop-bit count

Show hint

Check

Next

Idle

1

2

3

4

5

6

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13

14

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Core ideas

DIV

$\approx f_{sys} / \text{baud}$ (integer divide).

baud_tick

1-cycle enable every DIV clocks.

clk_div

Toggle every DIV $\rightarrow$ period 2·DIV.

OS

Sample clock: DIV $\approx f_{sys}/(\text{baud}\cdot\text{OS})$.

Baud divider starter

learn_uart — baud divider

Real RTL/TB practice

- In Track A, restate the divider formula in one sentence
- Draw a counter from zero to DIV minus one and mark where baud_tick goes high for one cycle
- Contrast that with a toggling clk_div output
- Optional: peek at UART RTL examples in the legacy combined materials and name where a
- This lab is divider literacy, not a full UART yet

Pitfalls to watch

- Do not confuse `baud_tick` with the serial line, it is an internal enable, one `sysclk` wide
- Toggle mode is not UART bit timing; its period is two times `DIV`, not `DIV`
- Integer division truncates, real designs pick `f_sys` and `baud` so the error stays acceptable
- With oversampling
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter DIV four pulse, step to a baud_tick at cnt three
- On paper, draw the counter wrap and one tick pulse
- When you are ready, take the short quiz, then continue to oversampling